Soccer Tournament Management System

Database Systems project

MSc Digital Transformation Management @UNIBO

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Disclaimer

Before starting the project, there are some disclaimers to be aware of:

- the sample data was created consistently with the project by ChatGPT; the query was written by hand with that data
- throughout the project, the soccer's games will be called MATCH in the design phase and GAME in the SQL queries to avoid misunderstandings with the MySQL keyword "match"
- the project is intentionally simple because it has an educational purpose, to understand how the whole process of designing, creating and filling a database works.

Requirements

The project focuses on the development of a database for a soccer tournament. In fact, it requires a database to efficiently manage data on the various league matches. The tournament is made up of several categories. Each team participates in a single category.

Each year, each team can be promoted or relegated. Every year, players can change teams based on their contracts.

The league is composed of multiple categories, each with its own name, year, and type.

These categories contain different teams, each with its own ID and name.

It is necessary to track the matches, each with a unique match ID, location, result, start time, and date; the participating teams and their respective players, each with a unique player ID, name, surname, goals scored, date of birth, and nationality; and the assigned referee, each with a unique referee ID, name, surname, date of birth, and nationality.

For each team, it is necessary to track the roster for each year composed of 24 players, each with a unique year, which identifies it along with the team and the number of goals scored.

This is linked to different player contracts and a specific category, both of which can change from year to year. Each contract has a unique contract ID that identifies it, a validity period, and the player's salary.

For each match it is necessary to record which category it belongs to, the teams that play it, the players who take part with their statistics for that match, i.e. goals scored, time played and any cautions, and finally the assigned referee.

Briefly, the database should facilitate the management of matches, teams, leagues, players and referees in an integrated way. The system should enable the tournament’s organizer to manage easily the data about new matches, changes of teams for the players, changes of categories for the teams, the insertion of new contract and the assignments of the referees.

Glossary of terms

Below is the brief glossary for the application. It should be useful to understand the meaning of the words used in the project and the other words linked to them.

For each term there is a brief description and the synonyms used inside the project.

TERM	DESCRIPTION	SYNONYM	LINK
PLAYER	Player included in a tournament	Sport performer, team member, competitor	Contract
CONTRACT	Contract between a player and a team which covers a specified number of years	Agreement, deal, commitment	Player, Season squad
SEASON SQUAD	Annual composition of player for a team	Season team	Team, Contract, Match, League
TEAM	Team included in a tournament	Group, sport association, society	Season Squad
MATCH	Match in a tournament	Football match, football challenge	Season Squad, Referee, League
LEAGUE	Different categories	Division, series	Season Squad, Match
REFEREE	Referee assigned to each match	Umpire, judge	Match

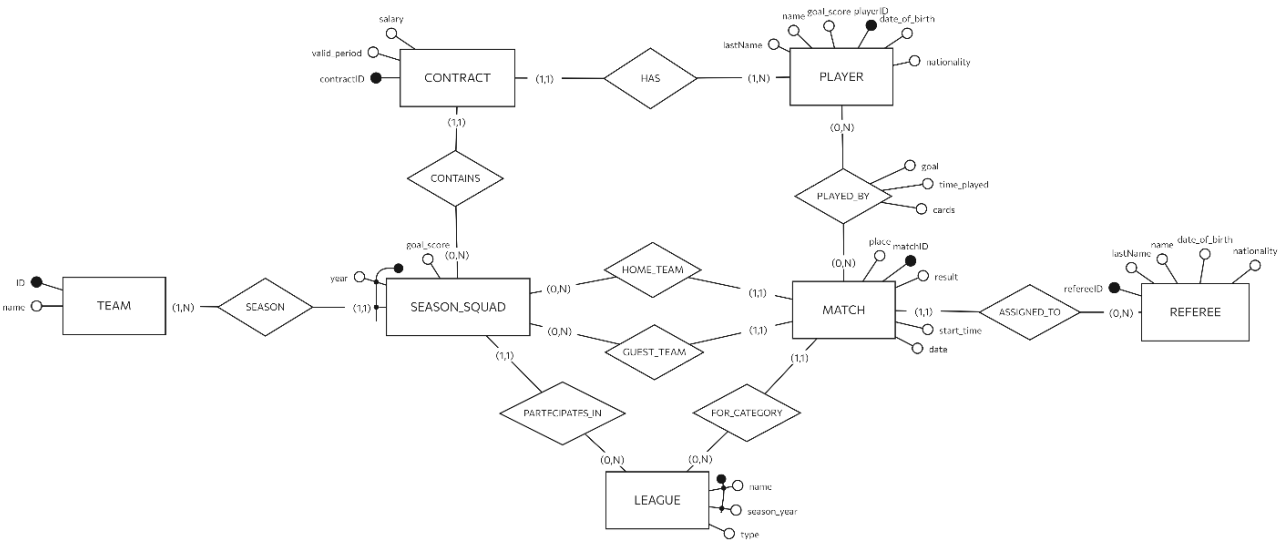
Table of Operations

Now we report the table of operations of our database, it focuses on the analysis of the main operation used in the management of the tournament ordered by their frequency. In this way we can show the eighty-twenty rule, to understand which are the main operations generating eighty percent of the work load.

Inside the table we report the expected frequency of any operations, and we specify if the operation is interactive (I) or batch (B).

NUMBER	OPERATION	AVERAGE FREQUENCY (avg of times a month)	TYPE
1	Insert a new player	2	I
2	Select players from a team	60	I
3	Find the player with the highest number of goals scored	40	B
4	Count matches without goals	20	B
5	Count the number of matches played by a team	10	I
6	Find all matches played in a certain day	10	I
7	Select all contracts with valid periods <= 1 year	5	B
8	Find player without goals scored	5	B
9	Select, for each team, the player with the highest number of goals scored, and order them	5	B
10	for each league, select the referee with the most matches refereed	5	B
11	Count the number of matches refereed by a certain referee	3	I
12	Insert a new referee	2	I
13	Select all players that have a more than 3 years contract and count per each valid period how many players have it	2	B
14	Find the salary of all players with more than 1 match played	1	I
15	Find the player with the highest salary	1	B

E/R Schema



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Non-Modellable Constraints:

- a contract cannot have a validity that overlaps with another contract of the same player, i.e. each player can have only one active contract at a given time, since the validity of the contracts of the same player must not overlap;
- the date of a match must fall within the period of the season to which it belongs;
- a season squad cannot play as the home team and as the guest one in the same match, and in the same season, a team cannot play as the home team twice against the same guest team.

Table of volumes

Concept	Type	Volume
Team	E	100
Season	R	500
Season Squad	E	500
Partecipates in	R	500
Contains	R	8000
Contract	E	8000
Has	R	8000
Player	E	2600
Played by	R	547200
Match	E	11400
Home player	R	11400
Guest player	R	11400
For category	R	11400
Assigned to	R	11400
League	E	6
Referee	E	120

Explanation

We started from the assumption that this soccer tournament was born in 2020, so it includes 5 full seasons. The number of different leagues in the tournament is 6, each team can have 24 players per year and plays 38 matches in a championship. Considering these assumptions, we have reached the following conclusions:

- a total of 100 **teams** was assumed;
- therefore, in the database 500 different **season_squad** has been registered;
- the number of **contracts** registered through the different seasons is 8000;
- there are 2600 **players** registered;
- a number of 6 **leagues** was assumed, each one with 20 **teams**.
- 11400 **matches** were played and registered in the database (5 seasons * 6 leagues * 380 matches). Each team plays against another team twice per season, so the number of matches per season in a league is 19 * 20 = 380.
- the number of **referees** registered is 120.

As for the relationships:

- *season* is a relationship between team (one to many) and season_squad (one_to_one), so the amount of this relationship is 500 (= 5 seasons * 100 teams);
- each **season_squad** participates in one and only one league, so the **participates_in** relationships are 500;
- *contains* relationship is the same quantity of contracts: 8000;
- *has* relationship is one-to-one with contracts, so, also this one is 8000;
- *played_by* is a relationship between player and match, we assumed 24 selected players per team in every match, so, the quantity of this relationship is 24 * 2 * 11400 matches = 547200;
- *home_player*, *guest_player*, *for_category* and *assigned_to* are all relationships one-to-one with match, so their quantities are all the same as the number of matches: 11400.

Restructuring phase

No restructuring is necessary to translate the schema, as there are no generalizations, hierarchies, compound or multi-valued attributes, or entities with multiple identifiers.

The only necessary change was to distinguish the two relationships, HOME and GUEST, within the MATCH table, more specifically their attributes, ID and year.

Translation phase

PLAYER (playerID, name, last_name, date_of_birth, nationality, goal_score)

TEAM (ID, name)

LEAGUE (season_year, name, type)

REFEREE (refereeID, name, last_name, date_of_birth, nationality)

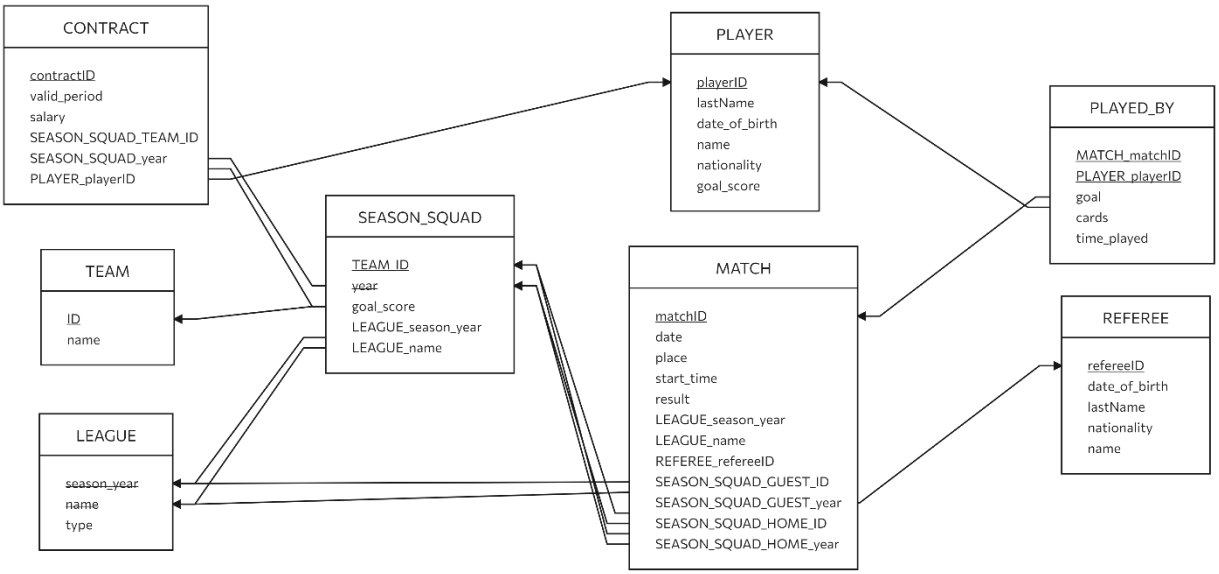
GAME (matchID, date, place, start_time, result, LEAGUE_season_year, LEAGUE_name, REFEREE_refereeID, SEASON_SQUAD_GUEST_ID, SEASON_SQUAD_GUEST_year, SEASON_SQUAD_HOME_ID, SEASON_SQUAD_HOME_year)

CONTRACT (contractID, valid_period, salary, SEASON_SQUAD _ID, SEASON_SQUAD_year, PLAYER_playerID)

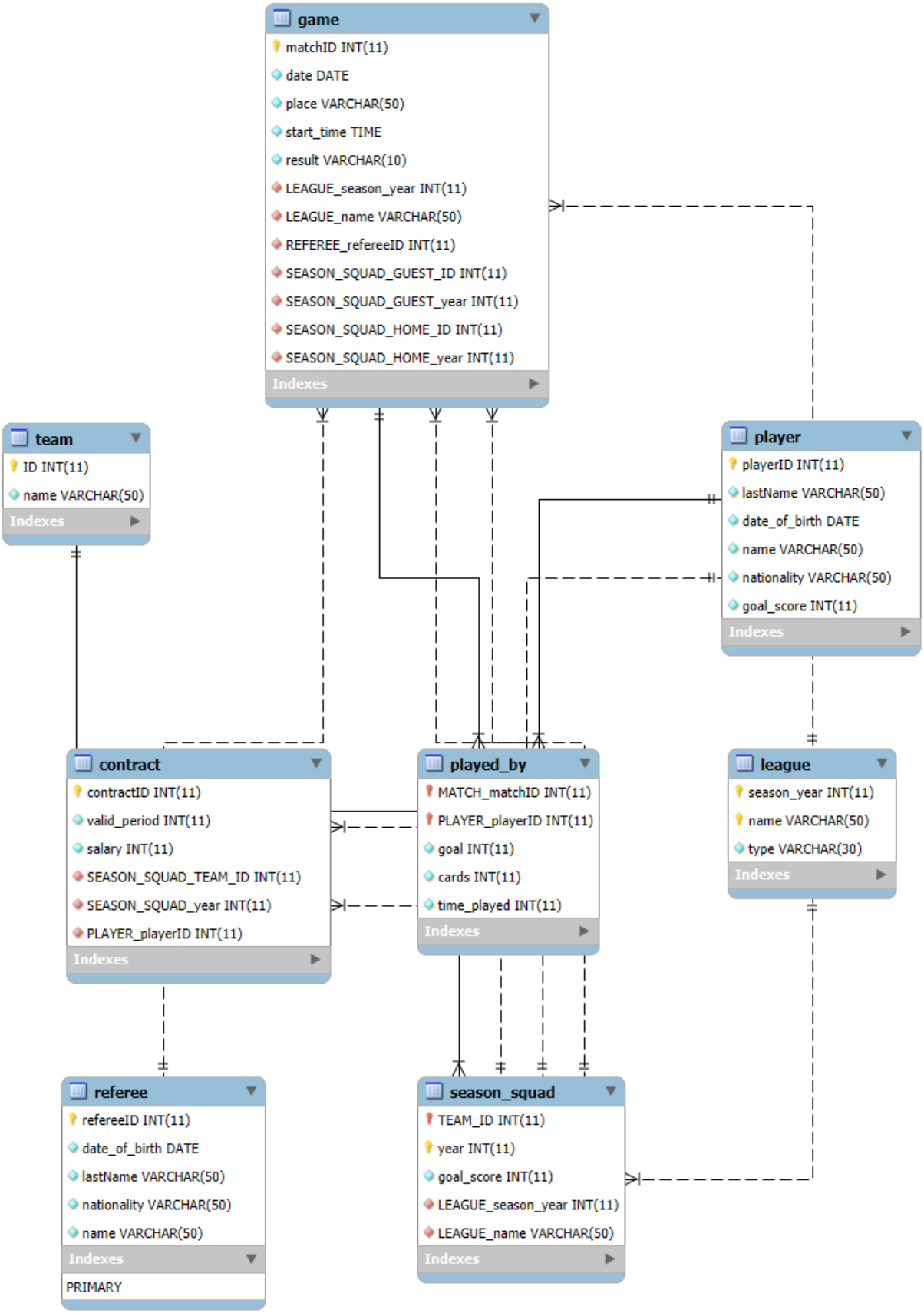
SEASON_SQUAD (teamID, year, goal_score, LEAGUE_season_year, LEAGUE_name)

PLAYED_BY (GAME_matchID, PLAYER_playerID, goal, cards, time_played)

Conceptual-to-logic model



Relational Schema



SQL Queries

1) Insert a new player

```sql
INSERT INTO PLAYER (playerID, name, last_name, date_of_birth, nationality, goal_score)
VALUES
  (121, 'Giuseppe', 'Rossi', '1999-12-04', 'Italy', 0)

SELECT *
FROM player
WHERE playerID = 121
```

Auto Commit | Manual

	playerID	name	last_name	date_of_birth	nationality	goal_score
1	121	Giuseppe	Rossi	1999-12-04	Italy	0

2) Select players from a team

```sql
SELECT p.playerID, p.name, p.last_name
FROM PLAYER p
JOIN CONTRACT c
    ON p.playerID = c.PLAYER_playerID
JOIN SEASON_SQUAD ss
    ON c.SEASON_SQUAD_ID = ss.teamID
    AND c.SEASON_SQUAD_year = ss.year
JOIN TEAM t
    ON ss.teamID = t.ID
WHERE t.name = 'Juventus';
```

Auto Commit | Manual

	playerID	name	last_name
1	101	Dusan	Vlahovic

3) Find the player with the highest number of goals scored

```sql
SELECT *
FROM player
ORDER BY goal_score DESC
LIMIT 1
```

Auto Commit | Manual

	playerID	name	last_name	date_of_birth	nationality	goal_score
1	102	Martinez	Lautaro	1997-08-22	Argentina	20

4) Count matches without goals

```
SELECT COUNT(*) AS matches_without_goals
FROM game
WHERE result = '0-0'
```

Auto Commit Manual

	matches_without_goals
1	1

5) Count the number of matches played by a team

```
SELECT COUNT(*) AS matches_played
FROM GAME g
JOIN TEAM t
    ON (g.SEASON_SQUAD_HOME_ID = t.ID
        OR g.SEASON_SQUAD_GUEST_ID = t.ID)
WHERE t.name = 'Juventus';
```

Auto Commit Manual

	matches_played
1	3

6) Find all matches played in a certain day

```
SELECT matchID, date, place, start_time, result, REFEREE_refereeID
FROM game
WHERE date = '2025-01-12';
```

Auto Commit Manual

	matchID	date	place	start_time	result	REFEREE_refereeID
1	3003	2025-01-12	Roma	15:00:00	2-1	203
2	3004	2025-01-12	Firenze	12:30:00	0-0	204

7) Select all contracts with valid period <= 1 year

```sql
SELECT c.contractID, p.name, p.last_name, t.name AS team,
       c.valid_period, c.salary
FROM CONTRACT c
JOIN PLAYER p
    ON c.PLAYER_playerID = p.playerID
JOIN TEAM t
    ON c.SEASON_SQUAD_ID = t.ID
WHERE c.valid_period <= 1;
```

Auto Commit Manual

	contractID	name	last_name	team	valid_period	salary
1	4007	Ciro	Immobile	Lazio	1	3500000

8) Find players without goal scored

```sql
SELECT *
FROM PLAYER
WHERE goal_score = 0;
```

Auto Commit Manual

	playerID	name	last_name	date_of_birth	nationality	goal_score
1	121	Giuseppe	Rossi	1999-12-04	Italy	0

9) Select, for each team, the player with the highest number of goals scored, and order them

```sql
SELECT t.name AS team, p.name, p.last_name, p.goal_score
FROM player P
JOIN contract c
    ON (p.playerID = c.PLAYER_playerID)
JOIN season_squad s
    ON (c.SEASON_SQUAD_ID = s.teamID)
JOIN team t
    ON (s.teamID = t.ID)
GROUP BY team
ORDER BY p.goal_score DESC;
```

Auto Commit Manual

	team	name	last_name	goal_score
1	Inter	Martinez	Lautaro	20
2	Napoli	Victor	Osimhen	18
3	Juventus	Dusan	Vlahovic	15
4	Bologna	Joshua	Zirkzee	13

10) For each league, select the referee with the most matches refereed

```sql
SELECT g.LEAGUE_name, r.refereeID, r.name, r.last_name, COUNT(*) AS matches_refereed
FROM GAME g
JOIN REFEREE r
    ON g.REFEREE_refereeID = r.refereeID
GROUP BY g.LEAGUE_name, r.refereeID
HAVING COUNT(*) >= ALL (
    SELECT COUNT(*) FROM GAME g2 WHERE g2.LEAGUE_name = g.LEAGUE_name
    GROUP BY g2.REFEREE_refereeID
);
```

Auto Commit Manual

	LEAGUE_name	refereeID	name	last_name	matches_refereed
1	Campionato Serie A 2024	201	Daniele	Orsato	3

11) Count the number of matches refereed by a certain referee

```sql
SELECT COUNT(g.matchID) AS No_of_matches, r.name, r.last_name
FROM game g
JOIN referee r
    ON (g.REFEREE_refereeID = r.refereeID)
WHERE g.REFEREE_refereeID = 203
GROUP BY r.name, r.last_name;
```

Auto Commit Manual

	No_of_matches	name	last_name
1	2	Anthony	Taylor

12) Insert a new referee

```sql
INSERT INTO REFEREE (refereeID, name, last_name, date_of_birth, nationality)
VALUES (211, 'Luca', 'Pairetto', '1984-04-01', 'Italia');

SELECT *
FROM REFEREE
WHERE refereeID = 211;
```

Auto Commit Manual

	refereeID	name	last_name	date_of_birth	nationality
1	211	Luca	Pairetto	1984-04-01	Italia

13) Select all players that have a more than 3 years contract and count per each valid period how many players have it

```
SELECT c.valid_period, COUNT(*) AS no_of_players
FROM contract c
JOIN player p
    ON (p.playerID = c.PLAYER_playerID)
GROUP BY c.valid_period
HAVING c.valid_period >= 3;
```

Auto Commit Manual

	valid_period	no_of_players
1	3	4
2	4	10
3	5	3

14) Find the salary of all players with more than 1 match played

```
SELECT p.playerID, c.salary
FROM player p
JOIN contract c
    ON (p.playerID = c.PLAYER_playerID)
JOIN played_by pb
    ON (p.playerID = pb.PLAYER_playerID)
GROUP BY p.playerID
HAVING COUNT(pb.GAME_matchID) > 1;
```

Auto Commit Manual

	playerID	salary
1	101	7000000
2	102	8500000
3	103	5000000
4	104	6000000

15) Find the player with the highest salary

```
SELECT p.name, p.last_name, c.salary
FROM contract c
JOIN player p
    ON (p.playerID = c.PLAYER_playerID)
ORDER BY 3 DESC
LIMIT 1;
```

Auto Commit Manual

	name	last_name	salary
1	Martinez	Lautaro	8500000